Syllabus Year: 2024-2025

CHRIST (Deemed to be University), Bengaluru - 560029

End Semester Examinations - November 2024

Bachelor of Computer Applications SEMESTER - III

Course Code: BCA201-3 **Max Marks: 100**

Course Name: COMPUTER NETWORKS **Duration: 03:00**

General Instructions:

- All rough work should be done in the answer script. Do not write or scribble in the question paper except your register number.
- Verify the Course code / course title and number of pages of questions in the question paper.
- Make sure your mobile phone is switched off and placed at the designated place.
- Malpractices will be viewed very seriously.
- Answers should be written on both sides of the paper in the answer book. No sheets should be detached from the answer book.
- Answers without the question numbers clearly indicated will not be valued. No page should be left blank in the middle of the answer book.

Course Outcome: Students will be able to:

CO1: Gain knowledge about networks, internal components and its functionality.

CO2: Understand the Data communication concepts. CO3: Trace and analyse any communication in network

CO4: Implement the networking concepts in real life scenarios.

Section A **2*10=20**

Answer all the Questions

1)	State important criteria the network must attain that makes the network better than others.	**2**	(CO1, L1)
2)	List different Packet Data Unit (PDU)'s used at different layers of TCP/IP Network Model.	**2**	(CO2, L1)
3)	Define guided and unguided transmission media.	**2**	(CO1, L1)
4)	Explain the advantages of optical fiber over twisted-pair and coaxial cable.	**2**	(CO3, L2)
5)	Define Piggy backing	**2**	(CO2, L1)
6)	Define is a burst error and its causes.	**2**	(CO3, L1)
7)	What are the four categories of standard ethernet?	**2**	(CO1, L1)
8)	List the IPv6 Header fields.	**2**	(CO3, L1)
9)	Define routing protocol.	**2**	(CO1, L1)
10)	List approaches of congestion control at timescale.	**2**	(CO2, L1)

Section B **5*6=30**

11)	Describe the differences between BUS and TREE Topology with diagrams.	**5**	(CO1, L1)
12) a)	Illustrate Interframe Space with respect to CSMA/CA.	**5**	(CO3, L2)
	OR		
b)	Discuss Code Division Multiple Access with an example.	**5**	(CO3, L2)
13)	Clearly state the types, advantages and disadvantages of Fiber optics.	**5**	(CO2, L1)

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14) a)	Elaborate congestion in a network with an example.	**5**	(CO3, L2)
	OR		
b)	Classify traditional cipher. Explain Transpositional Cipher with an example.	**5**	(CO3, L2)
15)	Examine the purpose of an NIC.	**5**	(CO4, L2)
16)	Illustrate classful addressing in IPv4.	**5**	(CO4, L2)

Section C **10*5=50**

17) a)	List the Network Topologies with a neat diagram. Also, point their advantages and disadvantages.	**10**	(CO1, L1)
	OR		
b)	State the functionality of DataLink, Network, Transport, Presentation and Applications Layers with clear diagrams.	**10**	(CO1, L1)
18) a)	List any five line coding schemes with valid examples and clear figures.	**10**	(CO2, L1)
	OR		
b)	Describe Amplitude, Frequency and Phase modulation with diagrams	**10**	(CO2, L1)
19) a)	Describe the protocol of Selective Repeat ARQ. Present all the cases of packet loss with neat diagrams.	**10**	(CO3, L1)
	OR		
b)	Describe flow control with the help of the following protocols. a) Stop and Wait. b) Automatic Repeat reQuest (ARQ).	**10**	(CO3, L1)
20)	Explain different techniques to improve QoS.	**10**	(CO3, L2)
21)	Draw TCP State diagram and examine the working of active connection termination.	**10**	(CO4, L3)

Revised Bloom's Taxonomy (RBT) Levels		
L1 - Remember	L2 - Understand	L3 - Apply
L4 - Analyse	L5 - Evaluate	L6 - Create

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